

Branch - CSE

Stream - 2022

Module :- 04

Polymers and Green fuels (8 hours)

Polymers

Introduction:-

Polymers are macromolecules having high molecular mass build up by linkage together of a large number of small molecules called monomers.

In Greek Poly means many, mers means units or part. The molecular mass of polymer is high and range from 10,000 to 1,00,000.

ex:- Polyethylene is a polymer formed by linkage of large numbers of ethylene.

Polymerization:- The chemical process in which the monomers are converted into polymers is called polymerization.

Degree of polymerization:- The number of times the unit is repeated in a homopolymer chain is referred to as the degree of polymerization.

Methods to find the molecular weight of polymer

Two methods

- By using number average molecular weight ($\bar{M}_n$)
- By using weight average molecular weight ($\bar{M}_w$)

* Number average molecular weight ($\bar{M}_n$)

$\bar{M}_n$ can be experimentally determined by using different techniques such as Osmometry, freezing point lowering, Boiling point elevation and X-ray diffraction

$\bar{M}_n$ is theoretically calculated by taking the ratio of the sum of the molecular weight contribution from the individual molecules to the number of molecules

Consider a sample of N polymer molecules containing n_1 molecules of molecular mass M_1 , n_2 molecules of molecular mass M_2 and so on ---

$$\begin{aligned}\text{Total number of molecules} &= n_1 + n_2 + n_3 + \dots + n_i \\ &= \sum_{i=1}^n n_i\end{aligned}$$

$$\text{Number fraction of } n_1 \text{ species} = \frac{n_1}{\sum_{i=1}^n n_i}$$

$$\begin{aligned}\text{Molecular weight contribution} \\ \text{from } n_1 \text{ species is} &= \frac{n_1 M_1}{\sum_{i=1}^n n_i}\end{aligned}$$

So the molecular weight contribution from all other species is

$$\bar{M}_n = \frac{n_1 M_1}{\sum_{i=1}^n n_i} + \frac{n_2 M_2}{\sum_{i=1}^n n_i} + \dots + \frac{n_i M_i}{\sum_{i=1}^n n_i}$$

$$\bar{M}_n = \frac{\sum_{i=1}^n n_i M_i}{\sum_{i=1}^n n_i}$$

$$\boxed{\bar{M}_n = \frac{\sum_{i=1}^n n_i M_i}{N}}$$

where $\sum_{i=1}^n n_i = N$.

* Weight Average Molecular weight ($\bar{M}_w$)

Weight average molecular weight is the molecular weight is the molecular weight contribution from all the species present in the polymer.

Total weight of the polymer = $W = \sum_{i=1}^n n_i M_i$

weight of n_1 species = $w_1 = n_1 M_1$

weight fraction of n_1 species = $\frac{n_1 M_1}{\sum_{i=1}^n n_i M_i}$

Molecular weight contribution from n_1 species = $\frac{w_1 \times M_1}{W}$

$$= \frac{n_1 M_1^2}{\sum_{i=1}^n n_i M_i}$$

Similarly molecular weight contribution from other species are

$$\frac{n_2 M_2^2}{\sum_{i=1}^n n_i M_i}, \quad \frac{n_3 M_3^2}{\sum_{i=1}^n n_i M_i} \quad \dots \quad \frac{n_i M_i^2}{\sum_{i=1}^n n_i M_i}$$

The weight average ~~aver~~ molecular weight is given by the expression

$$\bar{M}_w = \frac{n_1 M_1^2}{\sum_{i=1}^n n_i M_i} + \frac{n_2 M_2^2}{\sum_{i=1}^n n_i M_i} + \dots + \frac{n_i M_i^2}{\sum_{i=1}^n n_i M_i}$$

$$\boxed{\bar{M}_w = \frac{\sum_{i=1}^n n_i M_i^2}{\sum_{i=1}^n n_i M_i}}$$

Note :- Formula to calculate polydispersive index (PDI)

$$\frac{\bar{M}_w}{\bar{M}_n} > 1 \quad (\text{Heteropolymer})$$

$$\bar{M}_w = \bar{M}_n \quad (\text{Homopolymer})$$

Conducting polymers.

Conducting polymers are obtained by doping an Oxidizing or reducing agent into Organic polymers with Conjugated backbone consisting of alternating double and single Carbon-Carbon bonds.

Synthesis, Conducting mechanism and Commercial application of polyacetylene.

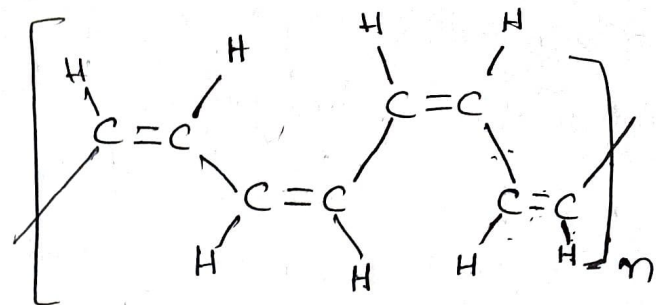
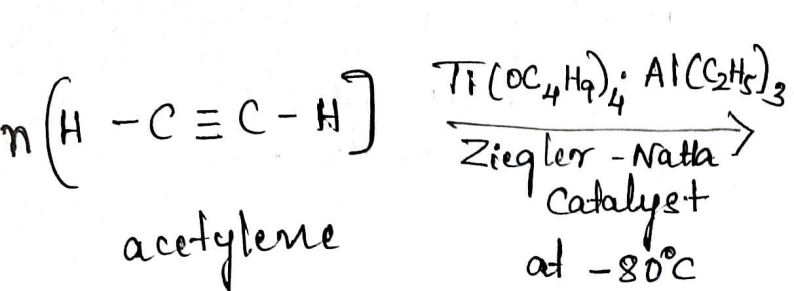
Synthesis

* A variety of methods have been developed to synthesize polyacetylene, from pure acetylene and other monomers.

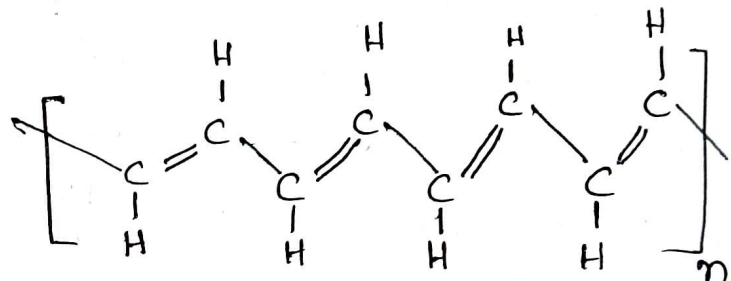
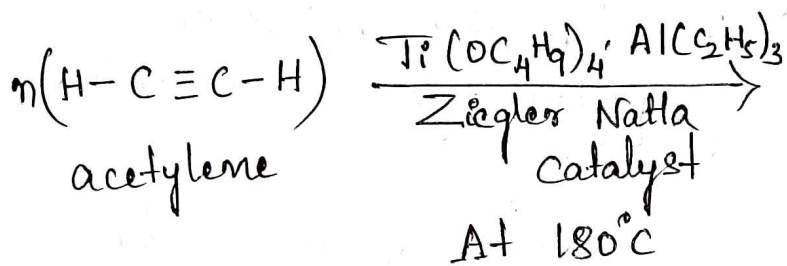
* One of the most common method uses a Ziegler-Natta Catalyst such as $\text{Et}_3\text{Al}/\text{Ti}(\text{OC}_3\text{H}_7)_4$ with gaseous acetylene. This method allows control over the structure and properties of final polymer by varying temperature and catalyst.

When acetylene gas is bubbled through heptane/toluene solvent containing Ziegler-Natta catalyst, polyacetylene solid film is formed at the gas-liquid interface.

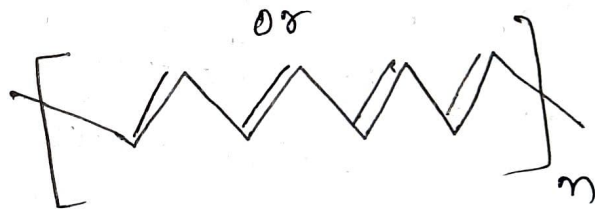
The controlled synthesis of each (Cis-polyacetylene and trans polyacetylene) of the polyacetylene can be achieved by changing the temperature at which the reaction is conducted.



cis polyacetylene (unstable)



trans polyacetylene (stable)



Conduction mechanism of polyacetylene.

p-type doping process to make the polyacetylene to conduct electricity.

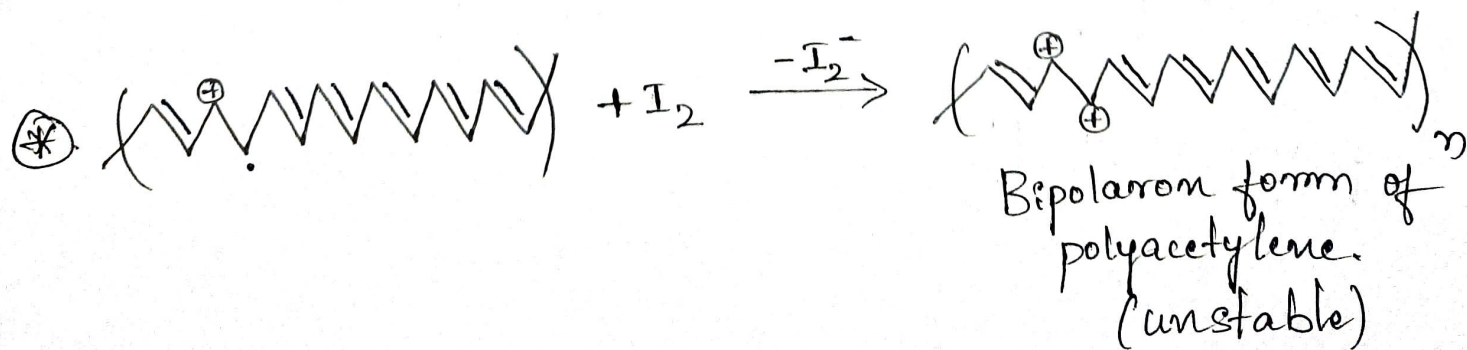
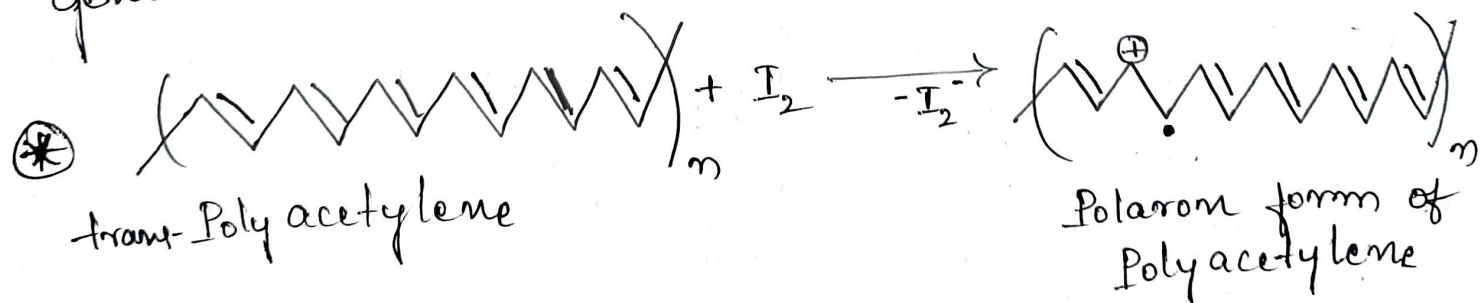
* When polyacetylene is doped with I_2 , it takes up electron from the polyacetylene, create a hole. then the polyacetylene start conducting the electricity. due to movement of holes in the ~~polymer~~ polyacetylene chain.

during this process I_2 takes up one electron from the polyacetylene create one hole and one free electron, this form of polyacetylene is called polaron form

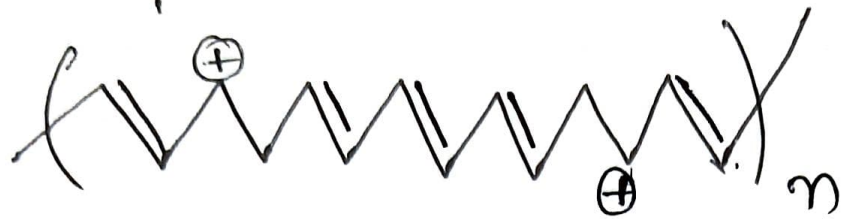
* When polaron form of polyacetylene treated with I_2 molecules, it takes up the another free electron from the polar form polyacetylene create a two positive charge in the polyacetylene chain.

These two positive charge present at the adjacent carbon atom. This form of polyacetylene is called bipolaron form

* Bipolaron form of polyacetylene is very unstable causes the electrons to move fastly. After some moment, the electrons undergoes re-arrangement and two cations are far away from each other that means not in the adjacent atom then generated a soliton form of polyacetylene



⊛ Finally come to the state of



Soliton form.

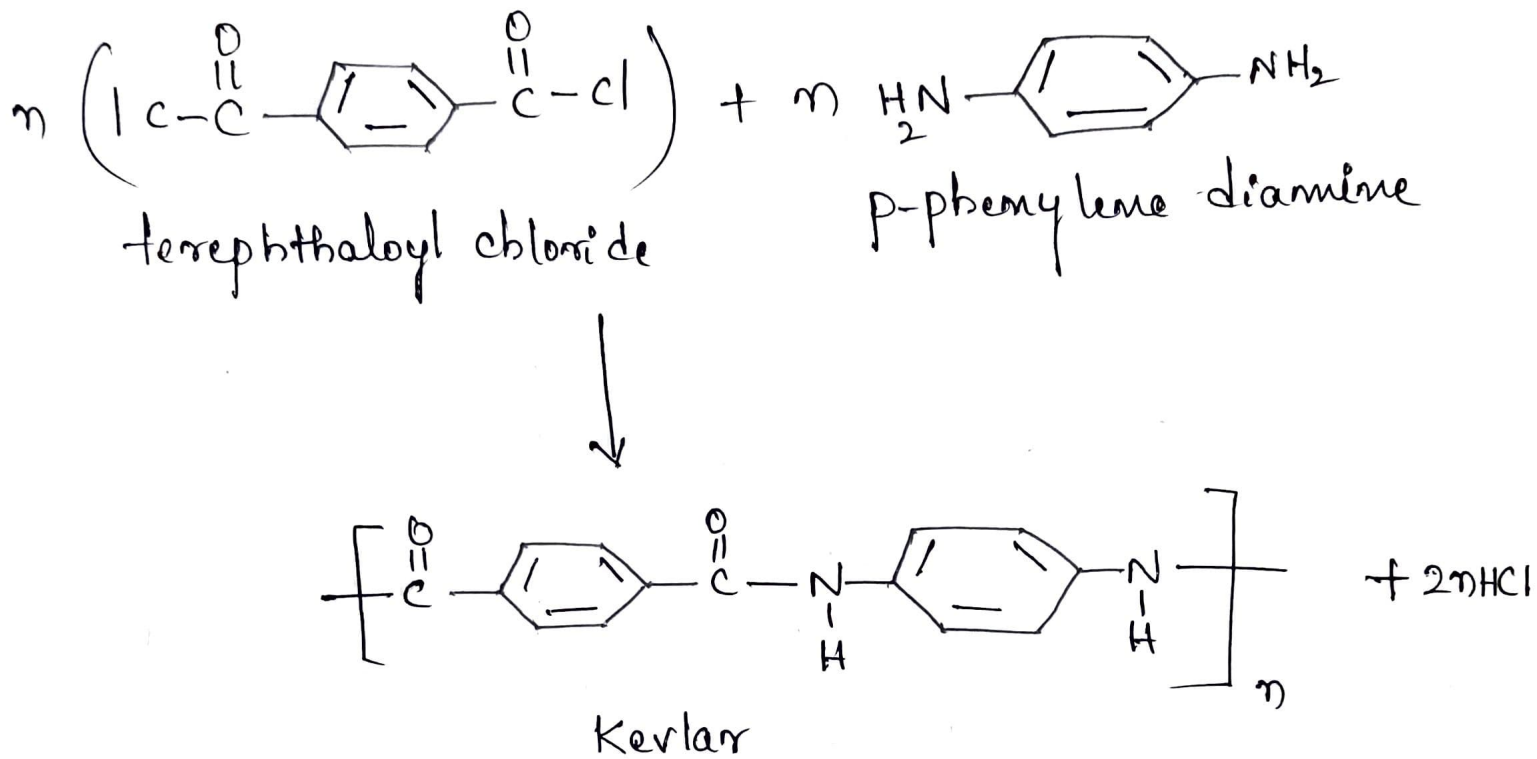
Commercial application of polyacetylene

- ⊛ Used in light-emitting diodes.
- ⊛ water purification devices
- ⊛ Hydrogen storage
- ⊛ Used as a biosensors.

Kevlar Fibre

Synthesis :- Kevlar is aromatic polyamide.

Kevlar is synthesized by the reaction of para-phenylene diamine and terephthaloyl chloride



Properties of Kevlar.

- Kevlar is crystalline, light weight and non-flammable.
- It has higher tensile and modulus than fiber glass.
- It has high abrasion resistance.
- It has good stiffness.

Application of Kevlar.

- It is used in light weight boat hulls, aircraft panels and race cars.
- It is used in bridge structures and bullet proof vests.
- It is used in puncture resistant bicycle tyres.

Synthesis of Graphene-Oxide.

Graphene Oxide was synthesized by Hummer method through oxidation of graphite. The stepwise preparation is given as follows:-

- Graphite flakes (2g) and NaNO_3 (2g) were mixed in 50 mL of H_2SO_4 (98%) in a volumetric flask kept under ice bath with continuous stirring.
- The mixture was stirred ~~at this~~ for 2 hours at this temperature and Potassium permanganate (6g) was added to the suspension very slowly.
- The rate of addition was carefully controlled to keep the reaction temperature lower than 15°C .
- The ice bath condition then removed and the mixture was stirred at 35°C until it become

- pasty brownish and kept under stirring for 2 days.
- It is then diluted with slow addition of 100mL water. The reaction temperature was rapidly increased to 98°C . with effervescence, and the colour changes to brown colour.
 - Further, add water then add 10mL H_2O_2 to terminate the reaction, this will appear by showing yellow colour.
 - The mixture finally washed by rinsing and centrifugation with 10% HCl and then with deionized water several time for purification.
 - After purification filter the mixture and dry under vacuum at room temperature, then finally Graphene Oxide was obtained as a powder.

Properties of Graphene Oxide.

1. Graphene Oxide is a brown colour powder.
2. Graphene Oxide is Hydrophilic in nature. So it is easily dispersable in water and other organic solvent. (due to the presence of oxygen functionality) So it form a stable colloid form.
3. Graphene Oxide contains both sp^2 and sp^3 -hybridized carbon atom, so it is more stable compare to Graphite.

4. Graphene Oxide poor conductor of electricity, but it is possible to increase the electricity by supplying the temperature increasing the temperature or by passing the light radiation through graphene oxide.
5. At room temperature Graphene Oxide acts as an insulator.
6. Graphene Oxide are highly hydrophilic in nature.

Application of Graphene Oxide

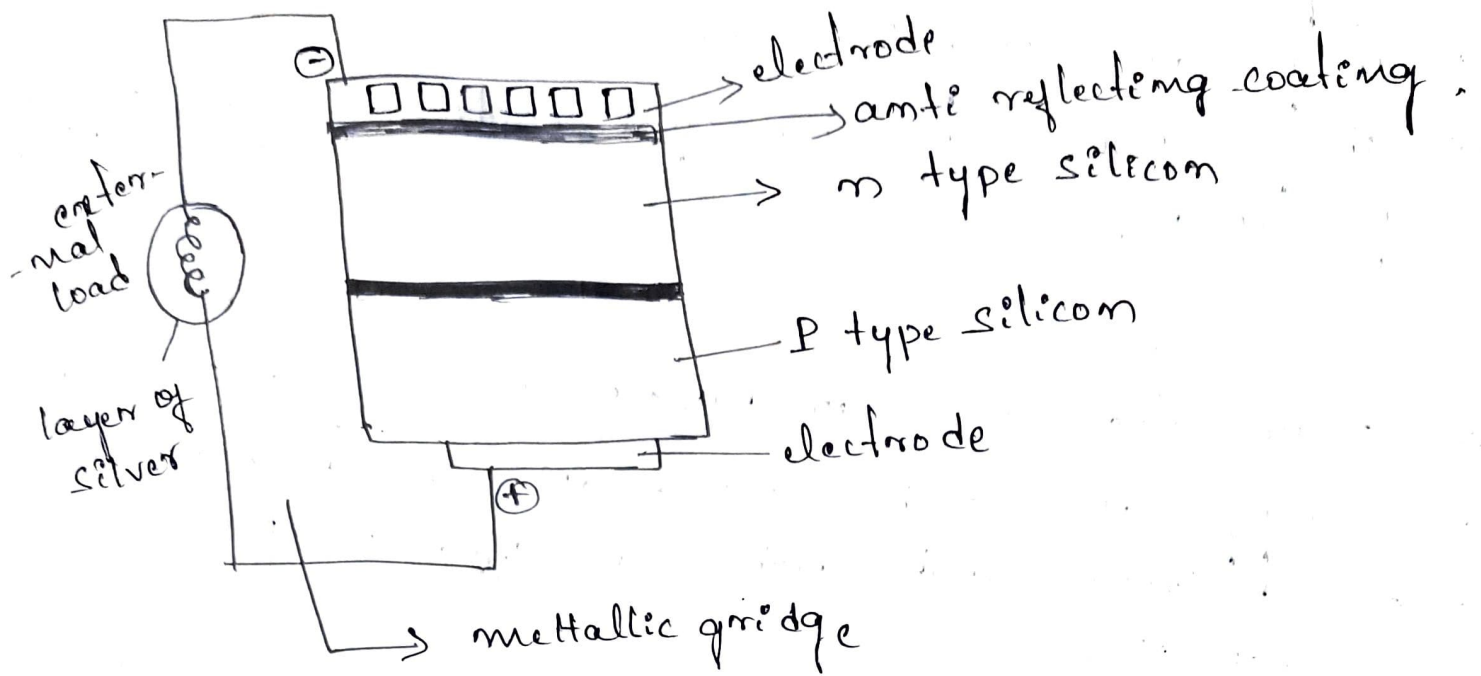
1. Graphene oxide is used in medicine ~~for~~ ^{treatment} cancer.
2. Graphene Oxide is used in drug delivery.
3. Graphene oxide is used to get the cellular imaging with the help of microscope.
4. Graphene oxides are used in chemical sensor.
5. Graphene oxides are used in biosensor.
6. Graphene oxides are used ~~is~~ as catalyst for some reaction.
7. Graphene Oxides are used in solar cells
8. Graphene Oxides are used in electronic devices.

Solar Cells or Photovoltaic cells

- Solar cells are the photovoltaic devices that convert the light energy (which comes from sun) into electrical energy.
- This device work on the principle of photovoltaic cell
- Photovoltaic device :- The generation of voltage across the p-n junction in a semiconductor due to the absorption of light radiation is called photovoltaic effect. The device based on this effect is called photovoltaic device.

Construction

- Solar cell consists of an ultra thin layer of phosphorous doped silicon (n-type) layer and boron doped silicon (p-type) layer. The two layers are sandwiched and hence there is formation of p-n junction.
- The surface is coated with anti-reflection coating to avoid the loss of incident light energy due to reflection.
- A proper metal contacts are made on the n-type and p-type side of the semiconductor for electrical connection.



Working

- When light radiation falls on the p-n junction diode electron-hole pairs are generated by the absorption of the radiation.
- The electrons are drifted towards the n-type and are collected at the n-type end.
- Holes are drifted towards the p-type and are collected at the p-type end.
- When the two ends are electrically connected through a conductor, there is a flow of electrons in the external circuit.
- Thus photoelectric current is produced and is available for use.

Advantages of Solar PV-cells

- It is clean and non-polluting.
- It is a renewable energy.
- Solar cells do not produce noise and they are totally silent.

- * They require very little maintenance.
- * They have long life time.
- * ~~They~~ There are no fuel costs or fuel supply problems.

Disadvantages of Solar Cells, or photovoltaic cell.

- * Solar cells or solar panels are very expensive.
- * Energy has not be stored as, like a batteries.
- * Air pollution and whether can affect the production of electricity.
- * They need large of land to produce more efficient power supply.
- * Less efficient and ~~cost~~ costly equipment.
- * Reliability depends on location.

Green fuels

Renewable hydrocarbon biofuels are called Green fuels.

(or)

Fuels produced from biomass sources through a variety of biological, thermal and chemical processes are called Green fuel.

Types of Green fuel.

1. Synthetic petrol

3. Power alcohol

2. Biodisels

4. Biogas

Green Hydrogen.

Hydrogen produced by the splitting of water by electrolysis process called Green hydrogen.

Production of hydrogen

Production of hydrogen can be done by two method.

- * electrolysis of water
- * solar water splitting.

Electrolysis of water.

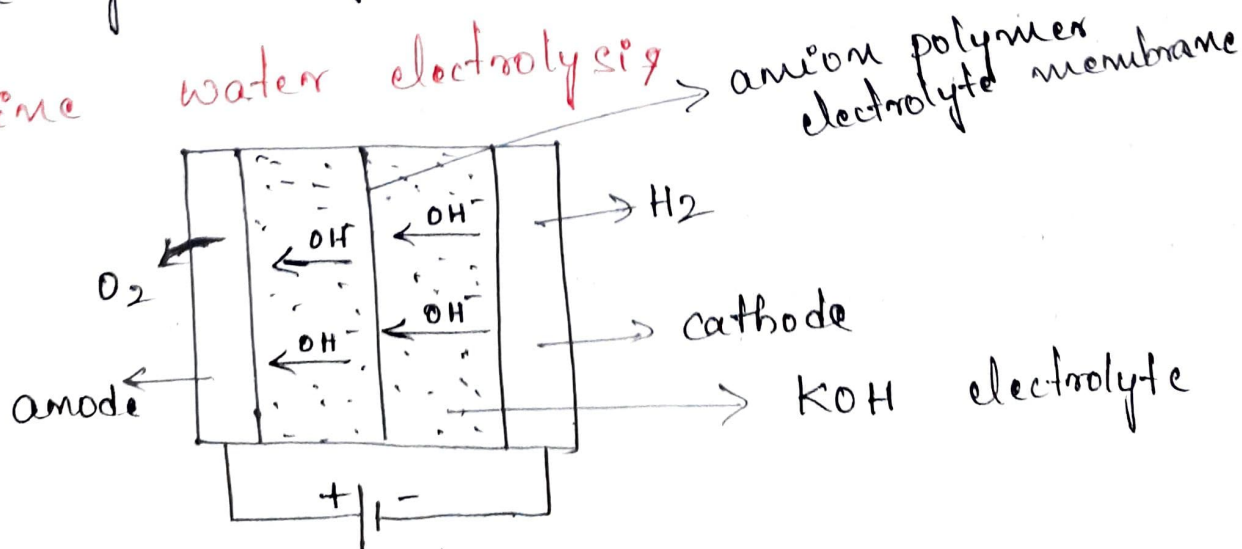
The decomposition of water into oxygen and hydrogen gas by passing an electric current through water is called as electrolysis of water.

There are three main ^{types of} electrolysis in water electrolysis

1. Alkaline electrolysis cell
2. Polymer electrolysis cell
3. ~~Solar~~ Solid oxide electrolysis cell.

Note:- 237.22 kJ/mol of energy and potential of 1.3V is required to split water into hydrogen + oxygen by electrolysis

Alkaline water electrolysis



The major components of alkaline water electrolysis cell are:-

* Anode:- Nickel metal dispersed on porous carbon is used as ~~cathode~~ anode.

At anode oxygen evolution reaction takes place

* Cathode:- Nickel metal dispersed on porous carbon is used as cathode.

At cathode hydrogen evolution reaction takes place.

* Electrolyte:- Aqueous potassium hydroxide (Aq. KOH)

* Separator:- Porous dense anion exchange membrane is used as separator.

It is a good ionic conductor for hydroxide ions.

It prevents the recombination of H_2 and O_2

Working:- ~~above~~

At cathode:-

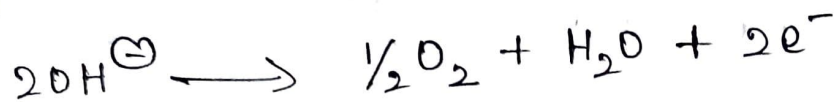
Water molecules are reduced to hydrogen and negatively charged hydroxide ion by passing the electricity from the external power supply



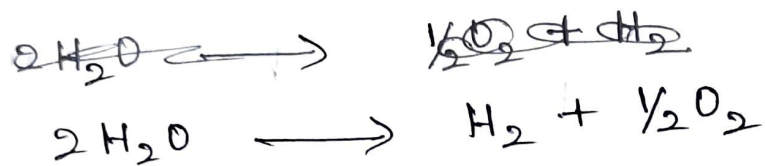
Hydroxide ions formed migrate to anode through electrolyte and separator.

At anode

Hydroxide ions are oxidized to oxygen and water



Over. all reaction



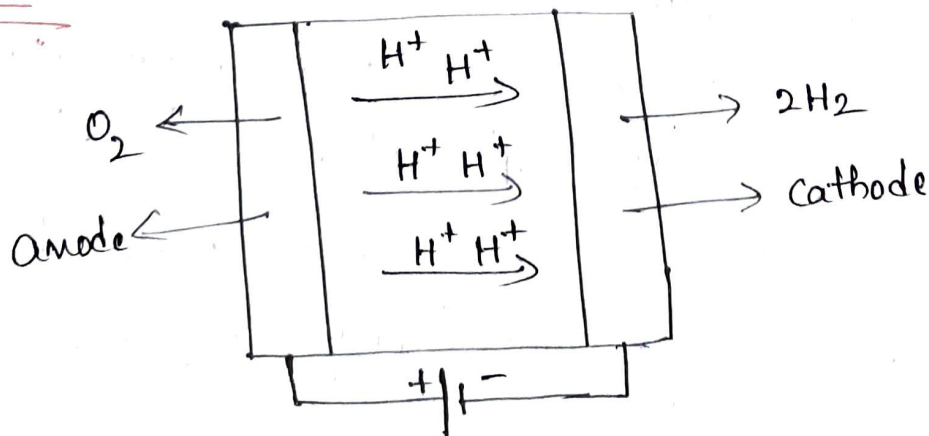
Advantages of alkaline electrolysis:-

- * Easier, cheaper and simple method for hydrogen production.
- * Electrodes are made up of cheaper nickel metal.
- * Pure carbon free hydrogen is obtained from this method.
- * Able to operate in low temperature.

Proton exchange membrane electrolysis

or
Polymer electrolyte ion exchange membrane electrolysis.

Construction



The major components in a polymer electrolyte membrane water electrolysis cell are

*Anode:- Iridium metal dispersed on porous carbon is used as anode.

This acts as a electrocatalyst for oxygen evolution reaction.

*Cathode:- Platinum coated on porous carbon is used as cathode

This acts as a electrocatalyst for hydrogen evolution reaction

* Tetrafluoroethylene

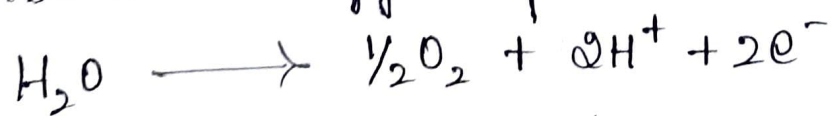
Solid polymer is used as both electrolyte as well as separator.

It is a good Ionic conductor of proton.

Working:- when electricity passes through external circuit.

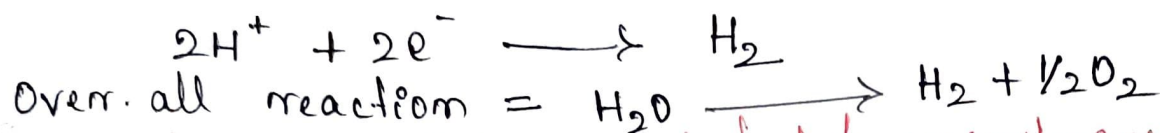
At anode:- Deionized water is circulated in the anodic chamber.

Deionized water undergo oxidation and liberates oxygen gas and hydrogen ions.



Hydrogen ions formed will migrate towards cathode through solid polymer electrolyte.

At cathode :- Hydrogen ions are reduced into molecular hydrogen (or) hydrogen gas.



Advantages of Polymer electrolyte membrane electrolysis

- * Use of polymer membrane avoids use of liquid acid as electrolyte
- * Chances of electrolyte leakage is prevented.
- * polymer membrane is chemically stable and non corrosive.

The main advantages of hydrogen as fuel are

- * Hydrogen is an ideal, highly efficient, renewable, and sustainable energy source
- * It is abundant from various sources i.e. from biomass or water.
- * Energy content of hydrogen is 122 kJ/g which is 2.75 times greater than by hydrogen fuel like petrol and diesel.
- * The combustion product is water
- * Used as chemical fuel to produce heat energy.
- * Has high storage capacity.
- * Used as fuel in fuel cell for production of electric current

